MILITARY SPECIFICATION

BENDING MACHINES, ROTARY HEAD, HORIZONTAL, HYDRAULICALLY OPERATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers horizontal, hydraulically operated, rotary head bending machines for making precision cold bends in pipe, tubing, and bar stock.

1.2 Classification. This specification covers bending machines of the following classes and of the sizes listed in table I. The class and size required for a specific procurement shall be as specified (see 6.2.1).

Class A - Bending machines with counterclockwise bending motion (left hand).

Class B - Bending machines with clockwise bending motion (right hand).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government publications. The following other Government publication forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and other Government publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B11.15 - Pipe, Tube, and Shape Bending Machines Safety Requirements for the Construction, Care, and Use.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, McLean, VA 22101.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The bending machine shall be new and one of the manufacturer's current models capable of horizontal, rotary bending of pipe, tubing, and bar stock. Bending shall be accomplished by wrapping the workpiece around a bending form or die. The design shall permit multiple bends with different degrees of bend angles in a workpiece without changing the setup. The machine design and construction shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one

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system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.2 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI System of Measurements.

3.2.2 Reclaimed materials. The bending machine may contain reclaimed materials provided such materials will not jeopardize its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 and ANSI B11.15 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

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3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the machine.

3.2.8 Environmental protection. The machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation (see 6.4).

3.2.9 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible. When specified (see 6.2.1), a centralized automatic lubrication system shall be provided.

3.2.10 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The bending machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

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3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28.

3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.8 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The bending machine covered herein shall consist essentially of the components described in 3.4.1 through 3.4.12.3. All components shall be capable of accommodating pipe and tubing of the sizes and capacities for the machine specified.

3.4.1 Base. The base shall be either cast iron, cast steel, or a fabricated steel weldment of a mass and weight that will support without permanent distortion all components of the bending machine while operating at the maximum rated capacity. Mounting bolt holes shall be provided for fastening the machine to the floor.

3.4.2 Nose. The nose assembly shall be a steel casting or weldment attached to the base. The nose shall be bored to accommodate precision antifriction bearings, or otherwise arranged to accept a cartridge type spindle housing which has been bored to accept precision antifriction bearings.

3.4.3 Spindle. The spindle shall be of precision ground alloy steel, sufficient in size and strength to withstand the maximum forces applied when the machine is operating at the maximum rated capacity. The spindle shall be mounted vertically on precision antifriction bearings mounted in the nose with provisions for adjustment. The die mounting boss or rotating swing arm shall be integral with the drive spindle.

3.4.4 Bending mechanisms. Unless otherwise specified, the machine shall be either of the integral bending form and clamp die type (see 3.4.4.1), or the swing arm type (see 3.4.4.2). When specified (see 6.2.1), the machine shall be of the type designated.

3.4.4.1 Bending form and clamp die type. When the bending form and clamp die type machine is furnished, the bending form and clamping die shall be an integral unit and hinged together. The clamping die shall be hydraulically operated with the control located conveniently to the operator at his normal work station. The clamping die shall clamp the pipe or tube to the bending form enclosing the full periphery of the workpiece. The clamping die shall be designed to provide self-alignment and uniform clamping pressure over the entire clamping area.

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3.4.4.2 Swing arm type. When a swing arm type machine is furnished the stationary and swing arms shall be steel castings, steel plate, or a combination thereof sufficient in strength to withstand the maximum force applied when the machine is operating at the maximum rated capacity. The arms shall be accurately machined and shall have slides mounted on replaceable wear strips or shall have provisions for lateral wear adjustment. The die mounting boss of the swing arm shall be fitted with keys and a locating ring for accurately positioning the bending dies. The clamping die assembly shall be mounted on the slide of the swing arm and shall operate by means of a hydraulic cylinder controlled linkage or a direct acting hydraulic cylinder. The clamping die assembly shall be designed to provide self-alignment and uniform clamping pressure over the entire clamping area.

3.4.5 Pressure die assembly. A pressure die assembly shall be provided to apply pressure during the bending operation. The pressure die assembly shall be mounted at right angles to the pipe or tube to be bent. The pressure die holder, slide, and ways shall be designed to provide and control pressure die force at the tangent point of the bend and on the vertical centerline of the pipe or tube to obtain maximum bend quality with minimum die deflection. The die holder shall accept pressure dies for all radii within the capacity of the machine. The pressure die assembly shall have means for lateral adjustment. The pressure die holder shall operate either by means of a direct acting hydraulic cylinder, or a hydraulic cylinder controlled linkage. Direct acting hydraulic pressure dies shall have a control, located conveniently to the operator's work station, and a gauge indicating the controlled die pressure.

3.4.6 Indexing carriage. When specified either a manually or power operated indexing carriage shall be furnished to facilitate loading the tube over the mandrel, moving the tube forward to the correct distance between bends (dbb) and rotating the tube to change plane of bend (pob). The particular type indexing carriage required shall be as specified (see 6.2.1). Machines supplied with an indexing carriage shall have a drop away clamp or other means to prevent interference with the bent pipe during the retraction cycle of the machine.

3.4.7 Mandrel extractor. A hydraulically operated mandrel extractor assembly sufficient in stroke to extract the entire ball assembly of a 3-ball type mandrel from the bent section of tube or pipe shall be furnished. The extractor shall have means for lateral adjustment to allow alignment of the mandrel for minimum and maximum bend die radii. Longitudinal adjustment shall be provided for mandrel tangent positioning. Provision shall be made to prevent accidental rotation of the mandrel during operation. Adjustable supports shall be provided to maintain and guide the mandrel and rod in alignment with the tube groove of the bend die.

3.4.8 Automatic mandrel lubricator. When specified (see 6.2.1), the bending machine shall be furnished with a mandrel lubricator which shall automatically lubricate the inside surfaces of the pipe or tube to minimize tool wear and galling. The lubricator shall accommodate all fluid type lub-

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ricants normally used in bending. The lubricator shall consist of a reservoir, piping, a pump or injector, and any other components necessary to provide the required amount of lubricant to the mandrel. When this option is specified, all mandrel rods furnished shall be hollow and of sufficient inside diameter to allow free passage of all fluid type lubricants normally used in bending.

3.4.9 Degree of bend selector. The degree of bend selector shall have preset adjustments for not less than eight different successive degree of bend angles. A calibrated scale shall be furnished to assist reading of bend angles in degrees. The amount of bending and the resulting bend angle shall be easily determined from the operator's work station. When specified (see 6.2.1), the degree of bend selector shall be the digital control type with means for adding spring back values.

3.4.10 Hydraulic system. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level. When specified (see 6.2.1), the hydraulic system shall conform to JIC H-1.

3.4.11 Tooling. When specified (see 6.2.1), tooling consisting of bending dies, clamp dies, pressure dies, wiper dies, and mandrels shall be furnished. When ordering pressure dies, and wiper dies the outside diameter, wall thickness, material and alloy of the tubing to be bent shall be specified. Orders for bending dies and clamp dies shall specify the outside diameter, wall thickness, material and alloy of the tubing to be formed, the centerline bend radius, and maximum bend angle required. Orders for mandrels shall specify outside diameter wall thickness, material and alloy of the tubing with which they will be used, the type mandrel, and the minimum centerline bend radius. Tooling ordered shall include all associated components and hardware to assure compatability with the machine specified.

3.4.12 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to NFPA 79. Each machine shall draw all of its electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating.

3.4.12.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor shall have a dripproof enclosure.

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3.4.12.2 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.12.3 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 120 volts.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than those specified in table I for the machine size specified.

3.6 Performance. Bending machines covered by this specification shall be capable of bending pipe and tubing of the sizes and capacities and to the radii specified in table I. The machine shall provide sufficient overbend to obtain a bend of 180 degrees in one cycle. The machine shall conform to the alignment tolerances listed in table II and the performance accuracies of table III.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. Unless otherwise specified, all equipment normally furnished by the manufacturer on his standard commercial machine shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.10 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.10.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions

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TABLE I. Sizes and capacities, classes A and B.

Size	Pipe, steel schedule 80 IPS (inches)	Tubing, stainless steel Y.P. to 60,000 psi (O.D. x wall thickness, inches)	Tubing, steel Y.P. to 40,000 psi (O.D. x wall thickness, inches)	Tubing, non-ferrous Y.P. to 25,000 psi (O.D. x wall thickness, inches)	Centerline radius (inches)	Tube length over mandrel (inches)	Bending speed, rpm (not more than)
1	3/4	1-1/8 x 0.065	1-3/8 x 0.065	1-1/2 x 0.065	2D to 8	102	27
2	2	2-1/2 x 0.109	3 x 0.072	3-1/2 x 0.120	2D to 12	102	10
3	3	4-1/2 x 0.083	5-1/2 x 0.095	5 x 0.156	2D to 24	150	4
4	4	4-1/2 x 0.165	6 x 0.120	6 x 0.250	2D to 24	150	3
5	6	6-3/4 x 0.118	7-1/2 x 0.250	7-1/2 x 0.250	2D to 36	150	1
6	8	9 x 0.093	12 x 0.125	12 x 0.125	2D to 42	150	1

NOTE: Requirements are not less than those shown except bending speed and centerline radius. Bending speed shall be not more than the figure shown. Centerline radius shall be from the smaller figure, or less, to the larger figure, or more.

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TABLE II. Alignment tolerances, classes A and B.

Die mounting ring in die boss concentric with rotation of spindle	0.005 inch TIR
Die boss surface parallel with stationary and swing arm ways and slides	0.015 inch TIR
Runout of spindle and die boss in 180°	0.005 inch TIR
Pressure die master bar to move parallel to die boss	0.002 inch TIR over 12 inch travel
Pressure die mounting surface perpendicular to die boss surface	90° ± 10 min.

TABLE III. Performance accuracies.

Bend angles	± 1/4°
Plane of bend	± 1/4°
Flattening (not more than)	4%
Wall thinning (not more than)	35%

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listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Hour meter. Unless otherwise specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.12 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the bending machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not

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authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturers designated indoor test area under the ambient temperature, the relative humidity, and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The bending machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Alignment tolerance test. The machine shall be tested in accordance with the contractor's standard test methods to determine that machine accuracy conforms to the alignment requirements of table II.

4.7.3 Performance tests. The machine shall be operated to bend two identical pieces of Schedule 80 steel pipe, capacity as designated in Table I, for the size machine being tested. The test pieces shall be bent on a centerline radius of twice the outside diameter in the following manner:

a. Make 180 degree bend at one end.
b. Rotate the test piece bending plane 90 degrees.

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c. Advance test piece and make a 90 degree bend.
d. Rotate test piece bending plane 90 degrees.
e. Move test piece forward and make another 180 degree bend.

Upon completion of the bends, one 180 degree bend of each test piece shall be cut at the center of the bend and measured for roundness and thinning of the wall. Two measurements of the inside diameter, at right angles to each other, shall be made. The two measurements shall vary not more than 4 percent. The wall thickness shall be measured to determine the amount of thinning. Thinning shall not exceed 35 percent. The completed test parts shall comply with the accuracies of table III.

4.7.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M- 18058.

6. NOTES

6.1 Intended use. The bending machines covered by this specification are intended for use in manufacturing and repair facilities for precision bending of pipe, tubing, and bar stock.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class and size required (see 1.2).

c. First article, if required (see 3.1).

d. If machine design and construction is required in a specific measurement system (US or SI), state required system (3.2.1).

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e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Dual calibrations (US and SI), if required (see 3.2.1.2).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Centralized automatic lubrication system, if required (see 3.2.9).

i. Painting, if different (see 3.3.5).

j. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

k. Specific type bending mechanism, specify type required (see 3.4.4).

l. If indexing carriage is required, state type (see 3.4.6).

m. Automatic mandrel lubricator, if required (see 3.4.8).

n. Digital degree of bend selector, if required (see 3.4.9).

o. Hydraulic system conforming to JIC H-1, if required (see 3.4.10).

p. Tooling, if different, fully describe (see 3.4.11).

q. Electrical system and input power, if different (see 3.4.12).

r. Motor enclosure, if different (see 3.4.12.1).

s. Size and capacity, if different (see 3.5).

t. Additional performance features, if required (see 3.6.1).

u. Standard equipment, if different (see 3.7).

v. Specify and fully describe optional equipment, if required (see 3.8).

w. Specify applicable repair parts, if required (see 3.9).

x. Lubrication chart or plate, if different (see 3.10.1).

y. Nameplate, if different (see 3.10.2).

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z. Hour meter, if different (see 3.11).

aa. First article inspection, if different (see 4.4).

bb. Quality conformance inspection, if different (see 4.5).

cc. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operators manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern the machine in its operating environment, such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. OSHA limits the total hazard level of these hazards in the environment and does not limit the hazard level of individual machine in the environment. The procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required because the guard configuration is dependent on the size and configuration of the workpieces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1 (g).

6.5 Subject term (key word) listing.

Bender, pipe, tube
Dies, bending
Floor mounting
Hydraulic
Machine
Mandrel
Metal fabricating
Pipe bender
Rotary bender
Tube bender

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6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - AL
Navy - SH
DLA- GS
Air Force - 84

User activities:
Navy - OS

Preparing activity:
DLA - IP

(Project 3441-0119)

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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